

HW2_23

2026-03-23

The data file collegetown contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in \$1000 units, PRICE, and total interior area in hundreds of square feet, SQFT.

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':  
##  
##   filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
data <- read.csv("collegetown.csv")  
  
str(data)
```

```
## 'data.frame':   500 obs. of  8 variables:  
## $ price      : num  96.5 106 114 119.9 122.7 ...  
## $ sqft       : num  16.8 17.5 18.4 29.1 14.7 ...  
## $ age        : int   10 10 9 9 9 8 10 9 7 10 ...  
## $ pool       : int    0 0 0 0 0 0 0 0 0 0 ...  
## $ fireplace: int    0 0 0 0 0 0 0 0 0 0 ...  
## $ close      : int    0 0 0 0 0 0 0 0 0 0 ...  
## $ twostory   : int    0 0 0 0 0 0 0 0 0 0 ...  
## $ occupied  : int    0 0 1 0 1 0 1 1 0 1 ...
```

```
summary(data)
```

```
##      price      sqft      age      pool
## Min.   : 50.0    Min.   :10.00   Min.   : 1.00   Min.   :0.000
## 1st Qu.: 157.9    1st Qu.:20.56   1st Qu.: 7.00   1st Qu.:0.000
## Median : 199.9    Median :25.55   Median : 9.00   Median :0.000
## Mean   : 250.2    Mean   :27.28   Mean   : 8.19   Mean   :0.064
## 3rd Qu.: 279.0    3rd Qu.:31.20   3rd Qu.:10.00   3rd Qu.:0.000
## Max.   :1370.0    Max.   :91.67   Max.   :11.00   Max.   :1.000
## fireplace      close      twostory      occupied
## Min.   :0.000    Min.   :0.000    Min.   :0.000    Min.   :0.00
## 1st Qu.:0.000    1st Qu.:0.000    1st Qu.:0.000    1st Qu.:0.00
## Median :1.000    Median :0.000    Median :0.000    Median :1.00
## Mean   :0.546    Mean   :0.378    Mean   :0.138    Mean   :0.56
## 3rd Qu.:1.000    3rd Qu.:1.000    3rd Qu.:0.000    3rd Qu.:1.00
## Max.   :1.000    Max.   :1.000    Max.   :1.000    Max.   :1.00
```

$$PRICE = \alpha_1 + \alpha_2 \times SQFT^2 + e$$

```
model <- lm(price ~ I(sqft^2), data = data)
summary(model)
```

```
##
## Call:
## lm(formula = price ~ I(sqft^2), data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -383.67  -48.39   -7.50   38.75  469.70
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  93.565854   6.072226   15.41  <2e-16 ***
## I(sqft^2)     0.184519   0.005256   35.11  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 92.08 on 498 degrees of freedom
## Multiple R-squared:  0.7122, Adjusted R-squared:  0.7117
## F-statistic: 1233 on 1 and 498 DF, p-value: < 2.2e-16
```

(a) Marginal Effect at 2000 sq ft

Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 \times SQFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p-value. What do you conclude?

- Step 1: 邊際效果公式

$$\frac{\partial E(PRICE)}{\partial SQFT} = 2\alpha_2 SQFT$$

$$2000sqft \Rightarrow SQFT = 20 \text{ ME} = \hat{\alpha}_2 \times 20$$

```
alpha2 <- coef(model)[2]
se_alpha2 <- summary(model)$coefficients[2,2]

ME_2000 <- 2 * alpha2 * 20
ME_2000
```

```
## I(sqft^2)
## 7.38076
```

- Step 2: Hypothesis Test

$$H_0 : 2\alpha_2(20) \leq 13$$

$$\Rightarrow H_0 : \alpha_2 \leq \frac{13}{40}$$

$$H_1 : 2\alpha_2(20) > 13$$

- t-statistics

$$t = \frac{\hat{\alpha}_2 - \alpha_{2,0}}{SE(\hat{\alpha}_2)}$$

```
alpha2_null <- 13 / 40

t_stat <- (alpha2 - alpha2_null) / se_alpha2
t_stat
```

```
## I(sqft^2)
## -26.72875
```

```
df <- nrow(data) - 2
t_crit <- qt(0.95, df)
t_crit
```

```
## [1] 1.647919
```

Since $|t| = 26.72875 > t_{0.95, n-2} = 1.647919$, we reject H_0 and conclude that the marginal effect on expected house price is greater than \$13000.

(b) Marginal Effect at 4000 sq ft

Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000.

Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p-value. What do you conclude?

$$4000sqft \rightarrow SQFT = 40$$

$$ME = 2\alpha_2 \times 40$$

```
ME_4000 <- 2 * alpha2 * 40
ME_4000
```

```
## I(sqft^2)
## 14.76152
```

- Hypothesis

$$H_0 : 2\alpha_2(40) \leq 13$$

$$\Rightarrow H_0 : \alpha_2 \leq \frac{13}{80}$$

$$H_1 : 2\alpha_2(40) > 13$$

```
alpha2_null_4000 <- 13 / 80

t_stat_4000 <- (alpha2 - alpha2_null_4000) / se_alpha2
p_value_4000 <- 1 - pt(t_stat_4000, df)

c(t_stat_4000, p_value_4000)
```

```
## I(sqft^2) I(sqft^2)
## 4.1894667148 0.0000165395
```

The test statistic is $t = 4.1894667148$.

Since $p\text{-value} = 0.0000165395 < 0.05$, we reject H_0 and conclude that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is significantly greater than \$13000.

(c) Expected Price at 2000 sq ft

Using the quadratic regression model in part (a), estimate the expected price

$$E(\text{PRICE}|\text{SQFT}) = \alpha_1 + \alpha_2 \text{SQFT}^2$$

for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.

- Standard Error

$$\text{Var}(\hat{y}_0) = \text{Var}(\hat{\alpha}_1) + x_0^2 \text{Var}(\hat{\alpha}_2) + 2x_0 \text{Cov}(\hat{\alpha}_1, \hat{\alpha}_2)$$

```
x0 <- 20

pred <- predict(model,
                newdata = data.frame(sqft = x0),
                interval = "confidence",
                level = 0.95)

pred
```

```
##           fit           lwr           upr  
## 1 167.3735 158.0481 176.6988
```

- The estimated expected amount of insurance is 167.3735.
- In repeated sampling, about 95% intervals constructed this way will contain the true value.

(d) Actual Sample Mean vs. Model

Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with SQFT = 20 compatible with the result in part (c)? Explain.

```
subset_20 <- data %>%  
  filter(sqft == 20)  
  
mean_price_20 <- mean(subset_20$price)  
mean_price_20
```

```
## [1] 163.3333
```

The sample average falls in the confidence interval in part (c), indicating that the actual sample mean is compatible with the result of the prediction model.